

z-test $\checkmark \rightarrow$ t-test

↓
 z-test when we perform any analysis using z-score.
 we require σ (population standard deviation)

z-test $\leftarrow \sigma \checkmark \quad \sigma \times \rightarrow$ t-test

$$z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \Rightarrow \boxed{t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}} \quad S = 7 \text{ Sample standard deviation}$$

Degree of freedom

$$\text{d.o.f.} \Rightarrow n-1$$

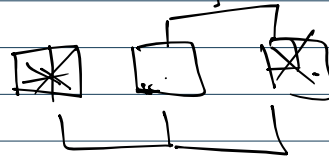
$$\Rightarrow 1^{\text{st}} = \frac{n}{3} \quad 1 = n = 3$$

$$2 = n-1 \quad 2 = n-1 = 2$$

$$= n-2 = 1$$

Ex:-

3 $\rightarrow$ 1st
 2nd



1st Person $\rightarrow$ 3 choices
 2nd $\rightarrow$ 2 choices
 3rd

T-distribution

t-test $\rightarrow H_0 \quad H_1$